Name: NIYONKURU Jean De La croix 03/04/2026
Reg N°: 223003235

MOBILE COMPUTING
Assignment Lab 6

Observations

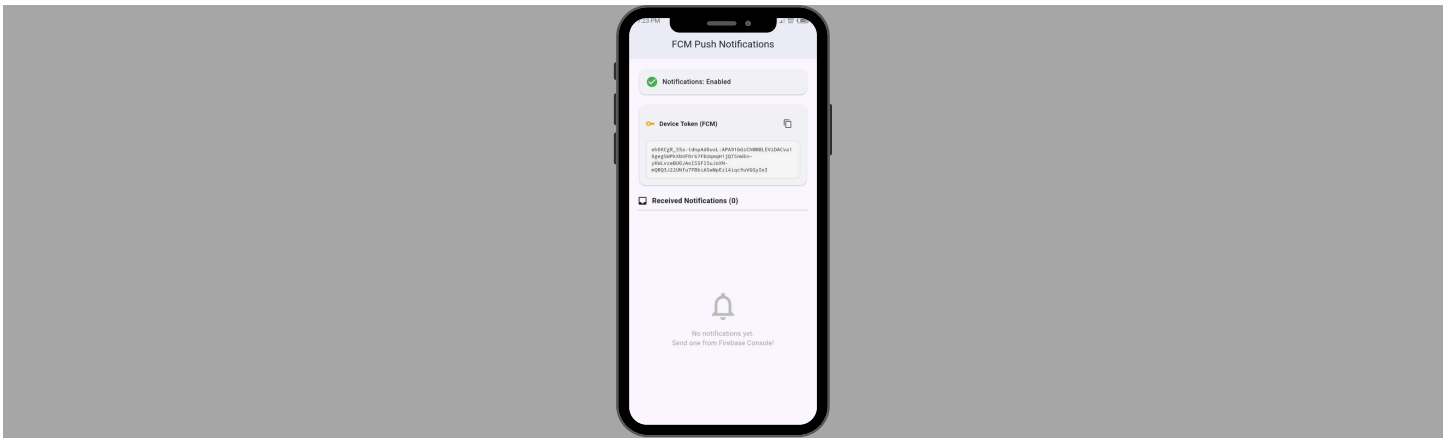
- Firebase Cloud Messaging (FCM) requires a real device; emulators don't receive notifications reliably.
- The app needs POST_NOTIFICATIONS permission on Android 13+; older versions allow them by default.
- Foreground notifications need flutter_local_notifications since FCM doesn't auto-show them.
- Background notifications are handled automatically by the system tray.
- The FCM token is unique per device and can change over time. If you reinstall the app, the token changes.
- The google-services.json file is essential. without it, Firebase won't initialize.
- Tapping a background notification reopens the app and we can read the message data from it.
- Testing on emulator was unreliable, a real phone was needed.

key steps followed

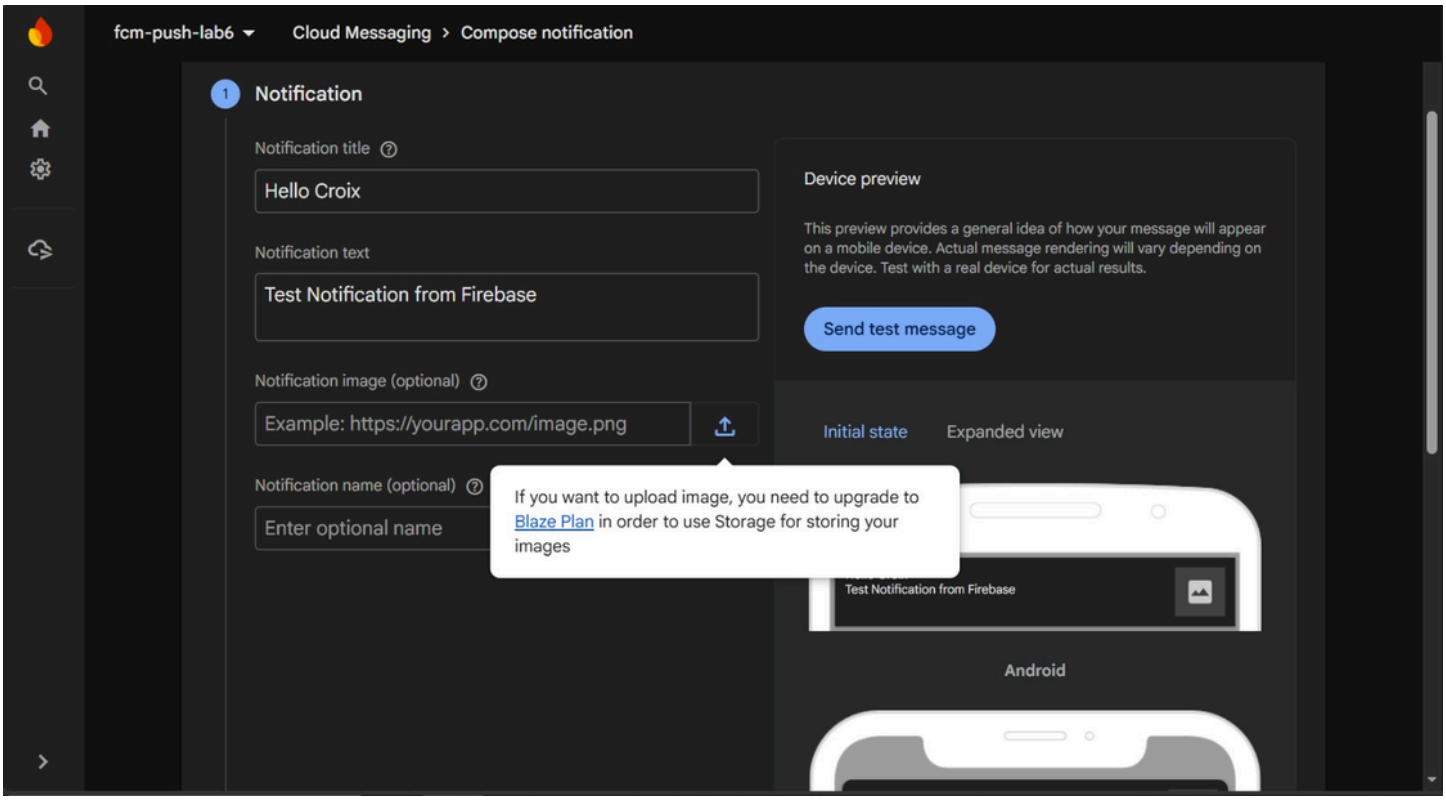
1. Created Flutter project and added firebase_core, firebase_messaging, flutter_local_notifications.
2. Created Firebase project and registered Android app
3. Installed Firebase CLI and ran flutterfire Configure to generate Config
4. Requested notification permission at runtime
5. Retrieved FCM device token and displayed it on screen
6. Set up listeners for foreground, background, and terminated states.

7. Sent test notification from Firebase Console using the device token.
8. Verified notification received in both states (foreground, terminated and background).

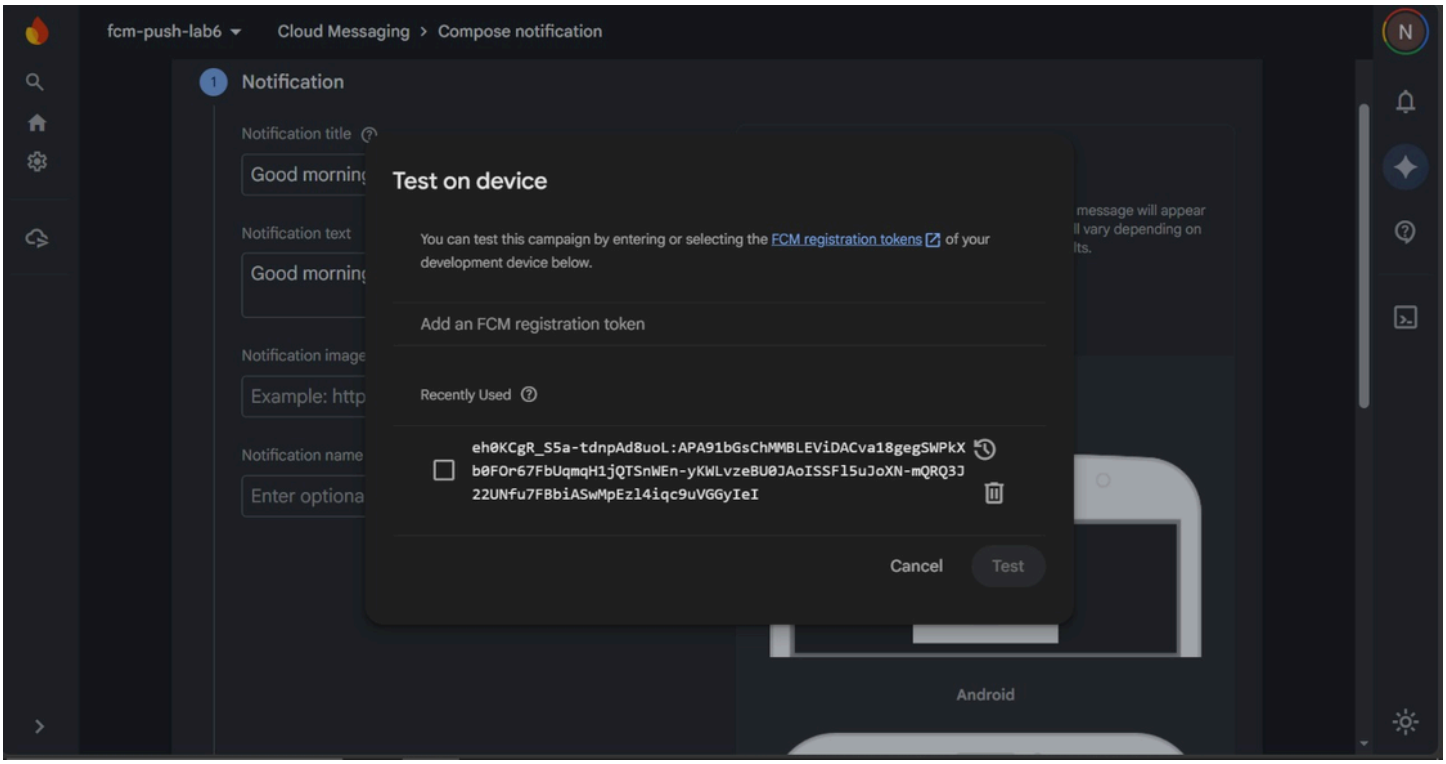
1. The very first moment we open the app on a new device, we see this beautiful screen with **device token id** without any notification received.



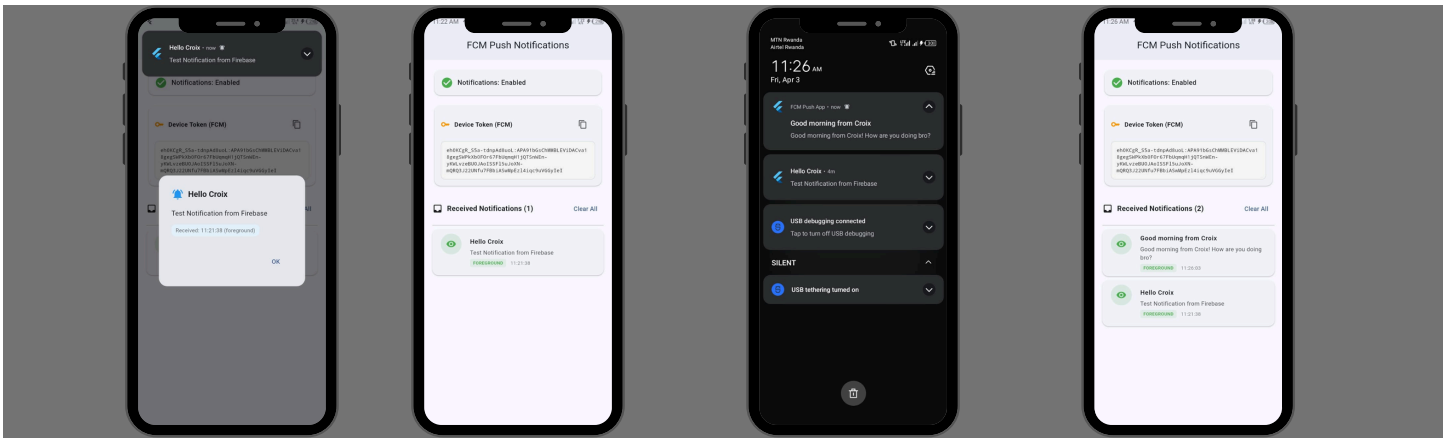
2. copied **device token id** and headed over to Firebase console, then to cloud messaging, and then to compose notification and typed in the testing message.



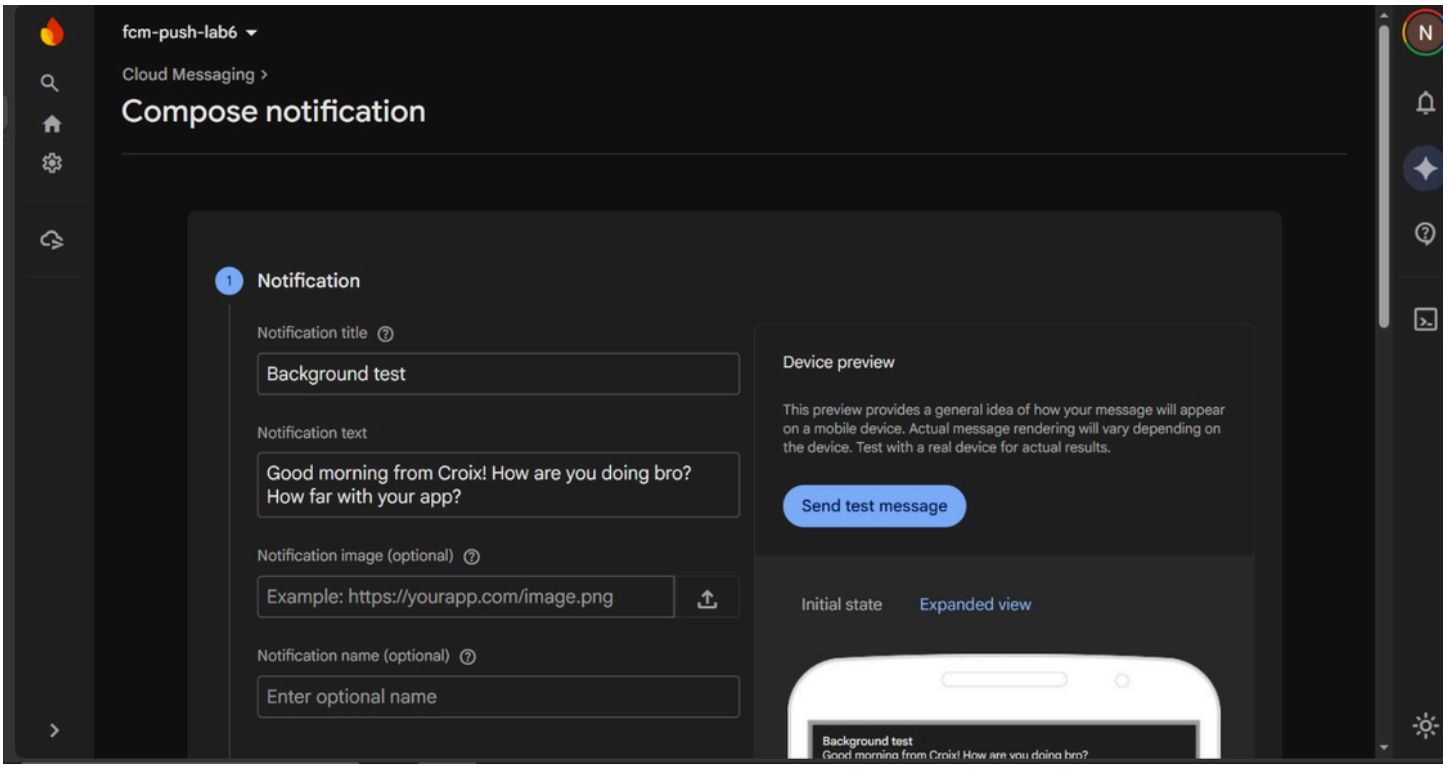
3. Clicked on the button that says Send text message and entered in **device token id** and saved it by checking it.

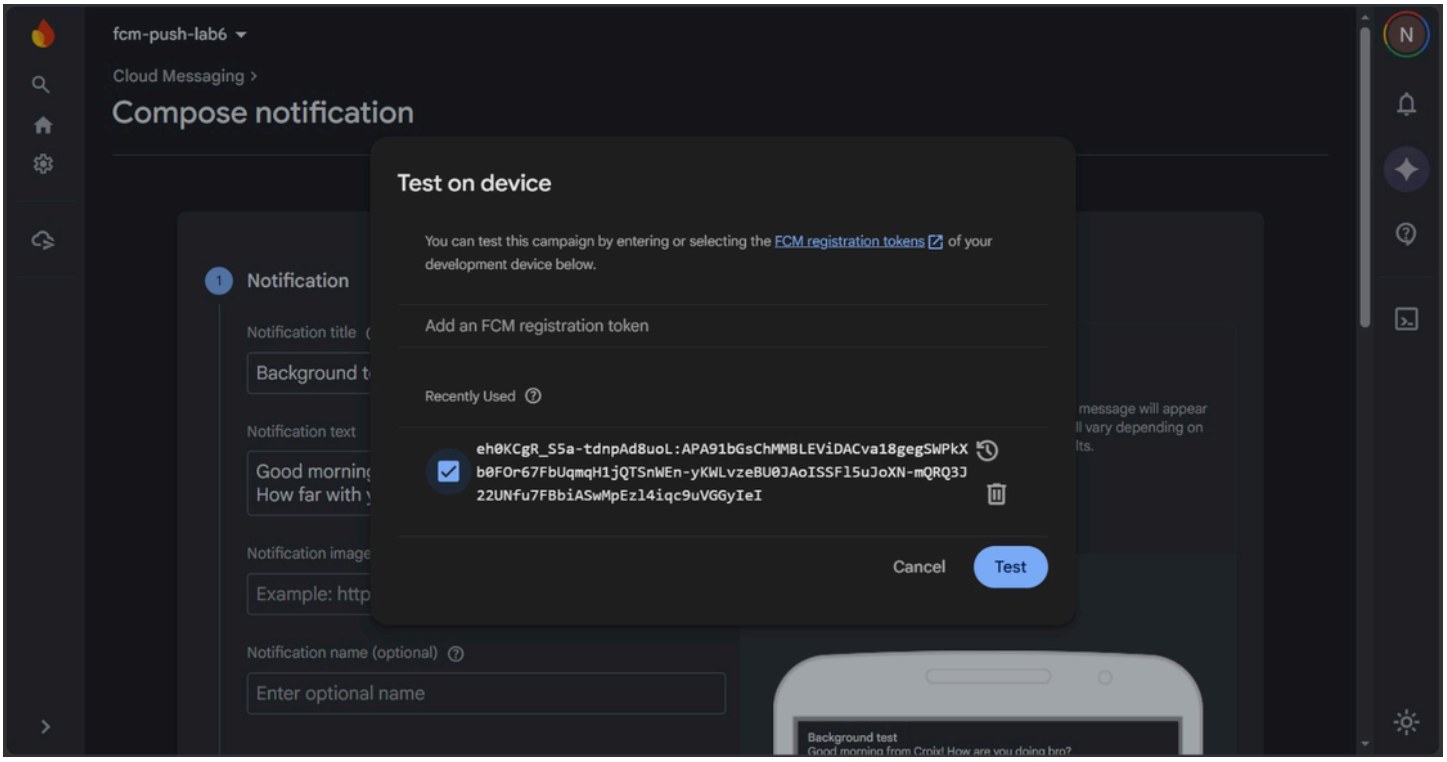


4. Received the message on my phone as screens show and it was in **foreground mode** because app was open.

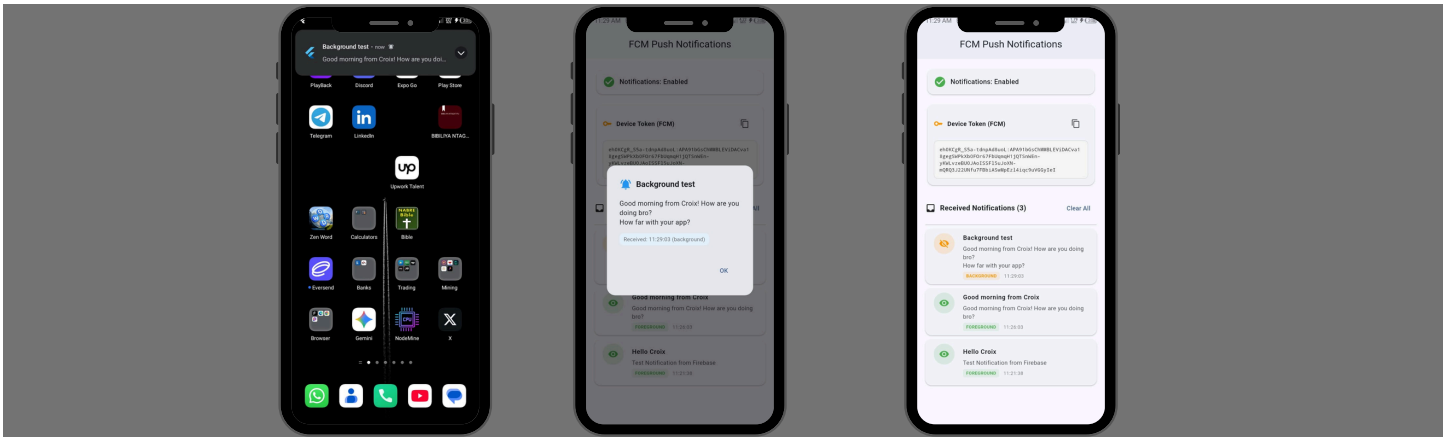


5. Tested **Background mode**

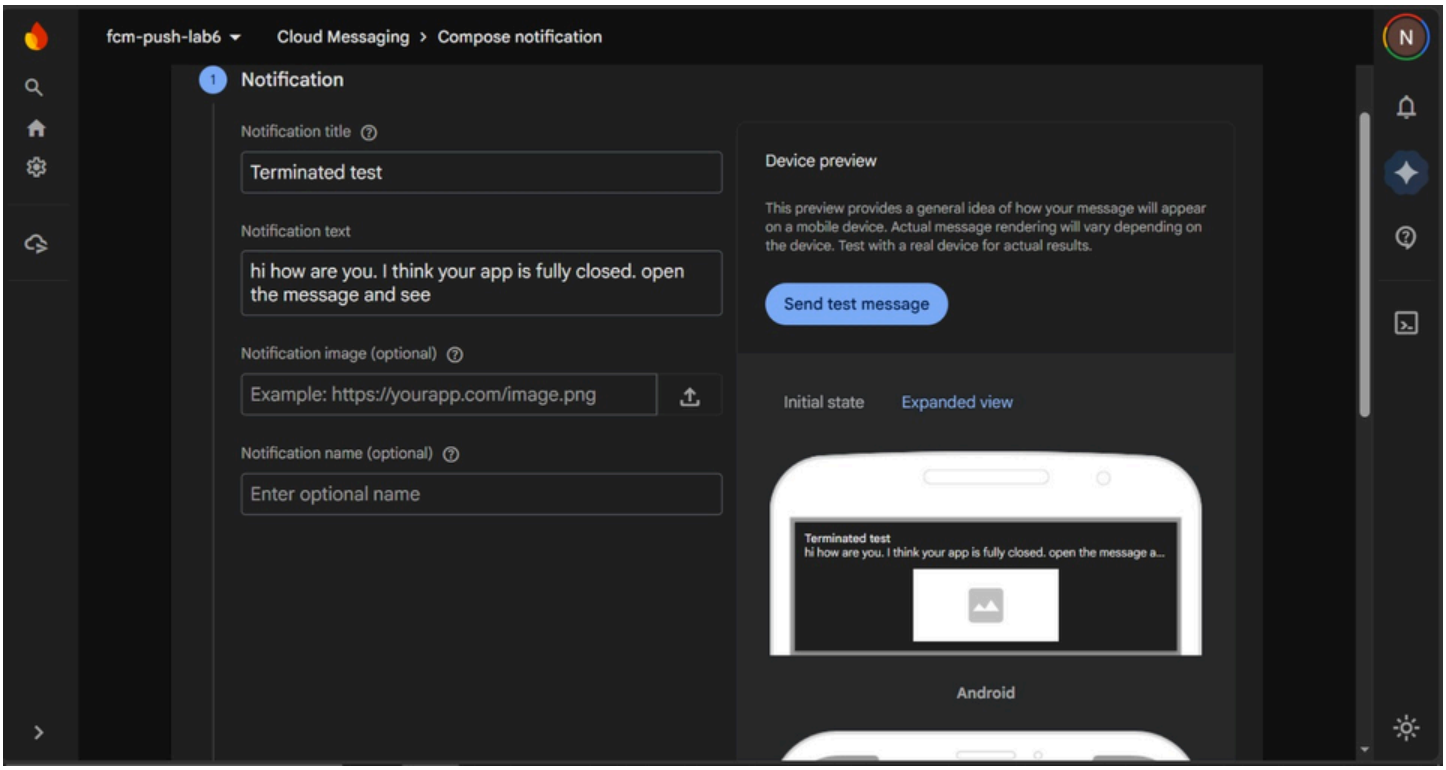


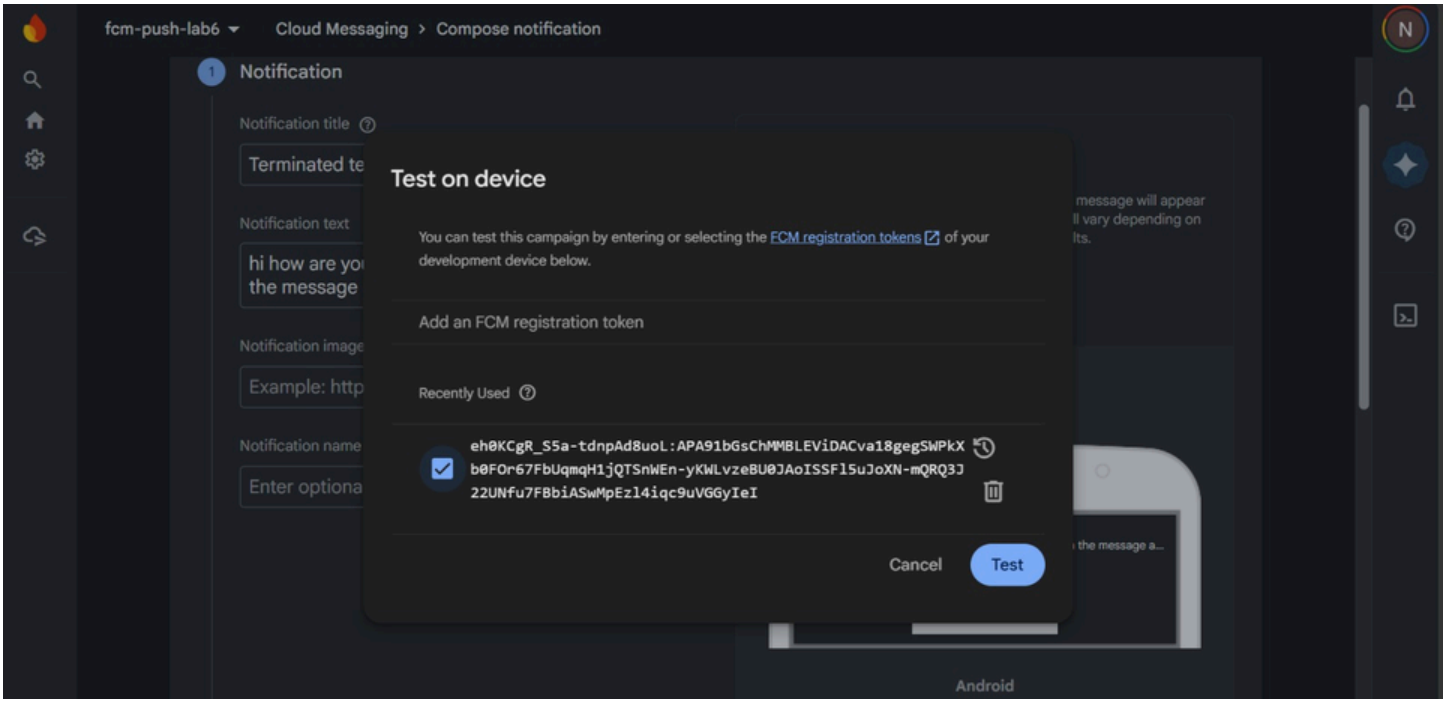


Received it on the device as a background message.

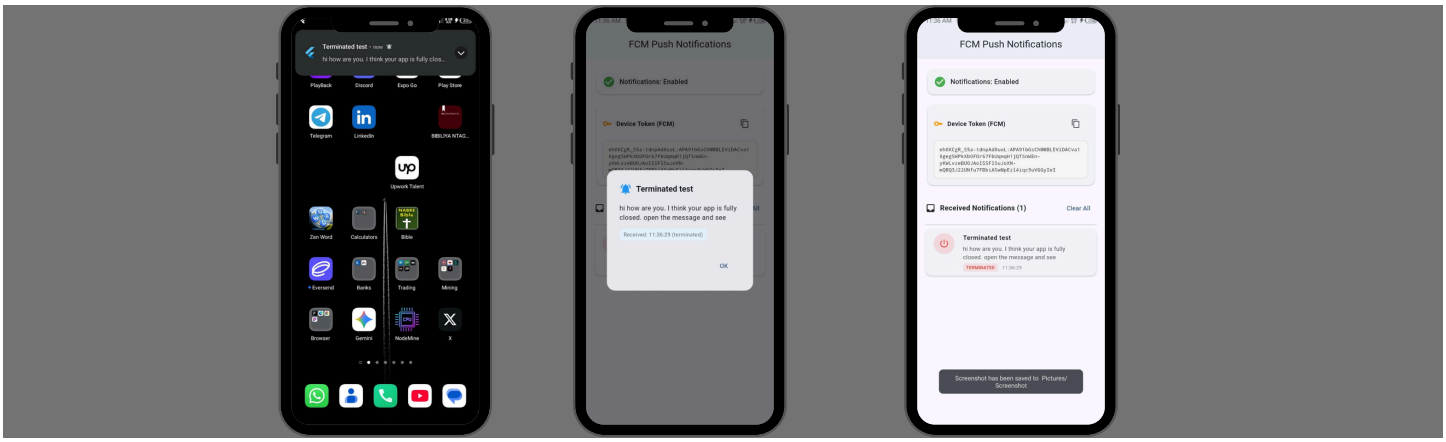
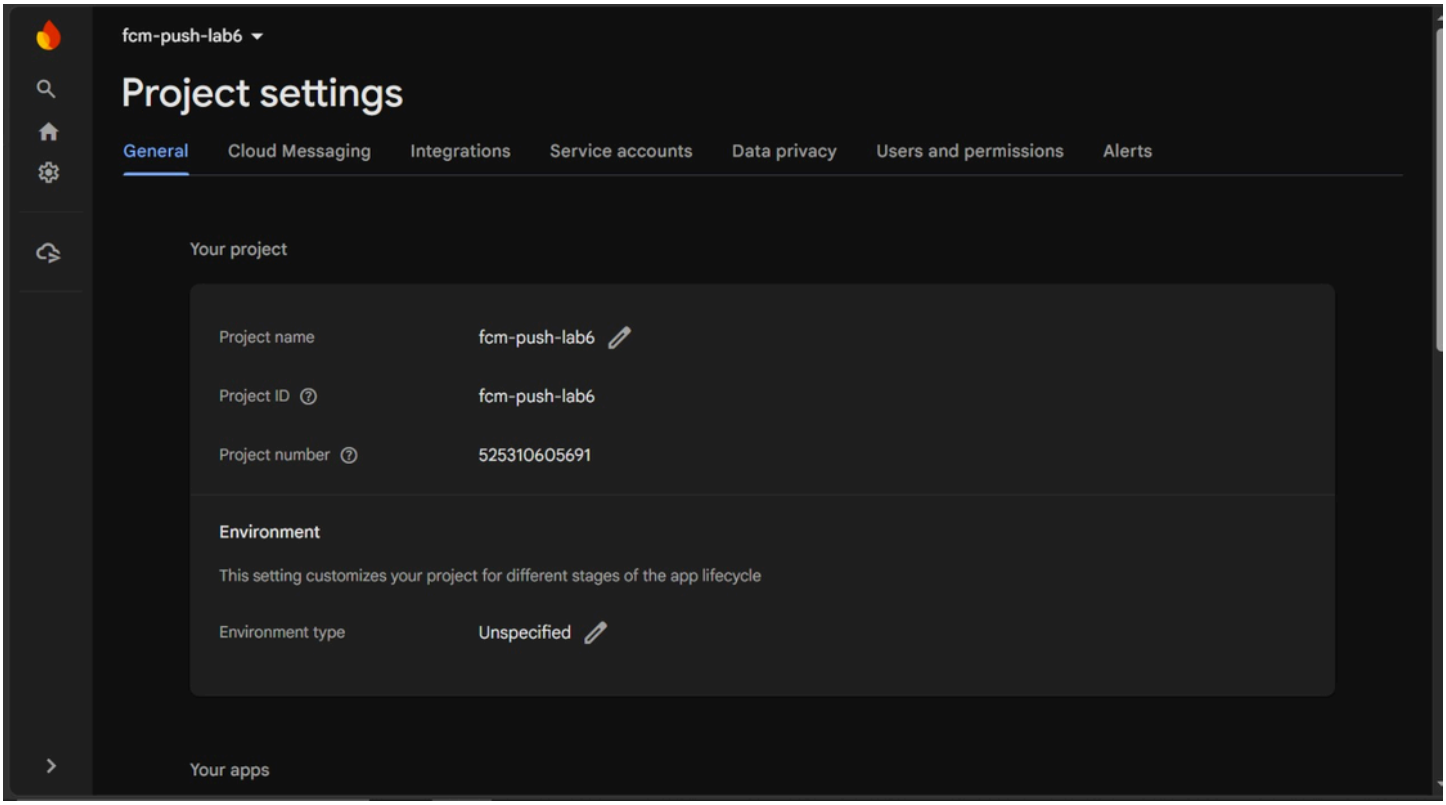


6. I closed the app completely to show how it behaves, and it showed **terminated state**





On the device, it opened in **terminated state**



Firebase settings: